RF_CHAIN_1

File: chain1.kicad_sch

RF_CHAIN_2

File: chain2.kicad_sch

RF_CHAIN_3

File: chain3.kicad_sch

RF_CHAIN_4

File: chain4.kicad_sch

LO_SYNTH

File: lo.kicad_sch

CAL_AND_TEST_IO

File: cal.kicad_sch

POWER_SEQUENCING

File: power.kicad_sch

MCU_SAFETY

File: mcu.kicad_sch

Sheet: /
File: openru_x6_tile.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 1/21

RF chain 1 (topology A): N310 ch0 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF1 <-> 4x ADTR1107 <-> 4 feeds.

BF_1

File: bf1.kicad_sch

FE_1

File: fe1.kicad_sch

CONV_1

File: conv1.kicad_sch

J101

C1_IF_N310

1

SMA IF1 (N310 ch0)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_1/
File: chain1.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

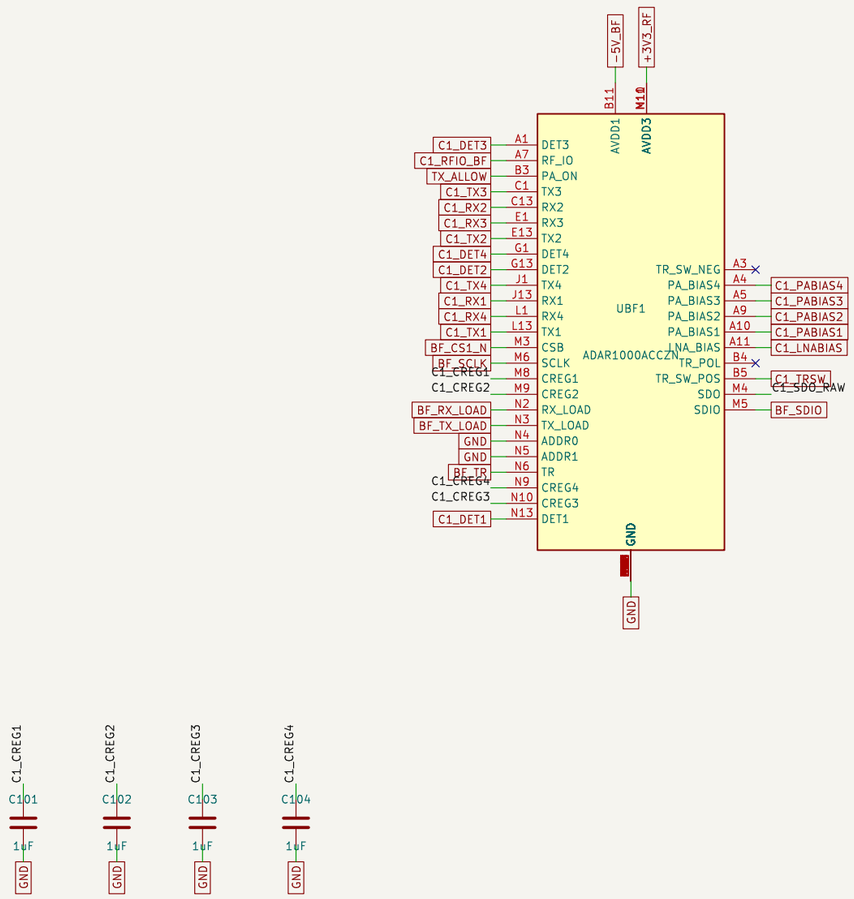
KiCad E.D.A. 9.0.9 | Id: 2/21

UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

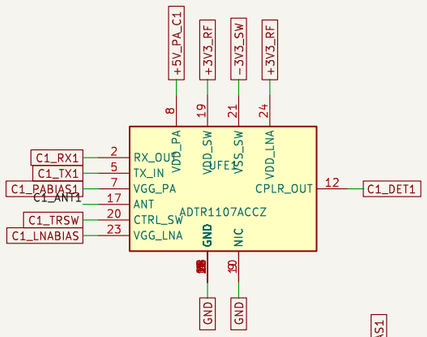
H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.

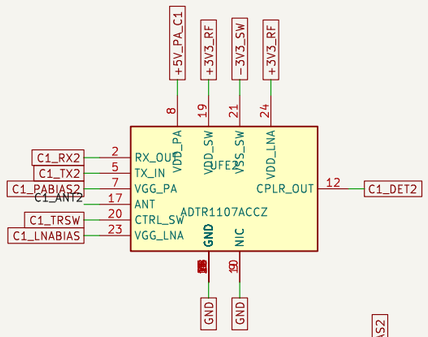


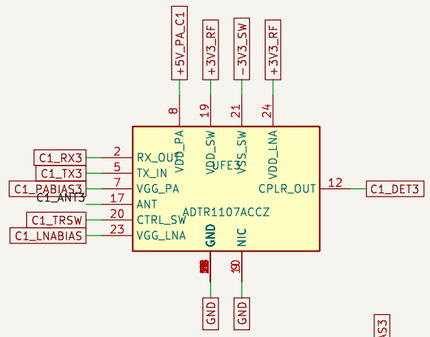
Sheet: /RF_CHAIN_1/BF_1/		
File: bf1.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 3/21

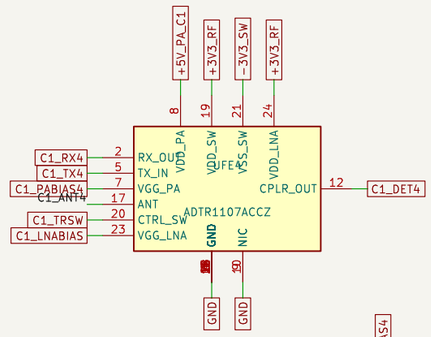
UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).







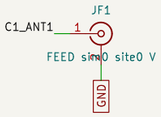


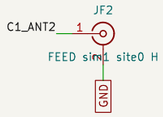


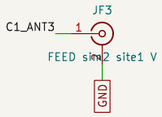


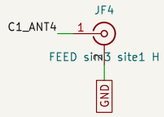


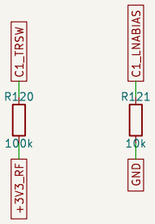












Sheet: /RF_CHAIN_1/FE_1/
File: fe1.kicad_sch

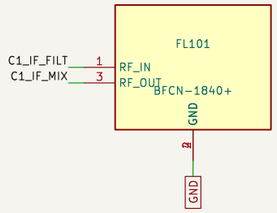
Title:

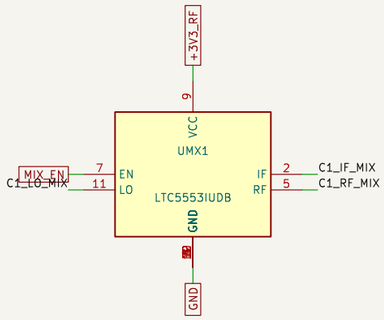
Size: A3 | Date: | **Rev:**

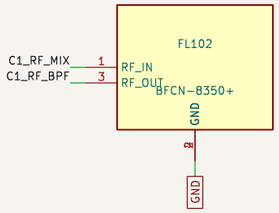
KiCad E.D.A. 9.0.9 | Id: 4/21

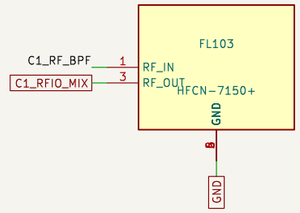
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_1/CONV_1/
File: conv1.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 5/21

RF chain 2 (topology A): N310 ch1 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF2 <-> 4x ADTR1107 <-> 4 feeds.

BF_2

File: bf2.kicad_sch

FE_2

File: fe2.kicad_sch

CONV_2

File: conv2.kicad_sch

J201

C2_IF_N310

1

SMA IF2 (N310 ch1)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_2/
File: chain2.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

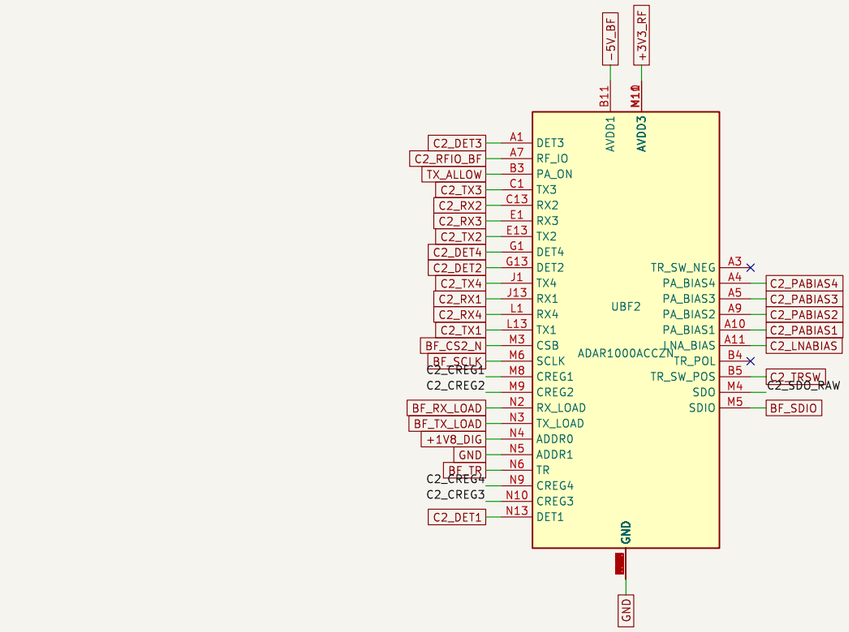
KiCad E.D.A. 9.0.9 | Id: 6/21

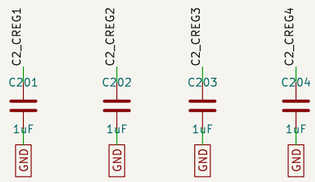
UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.



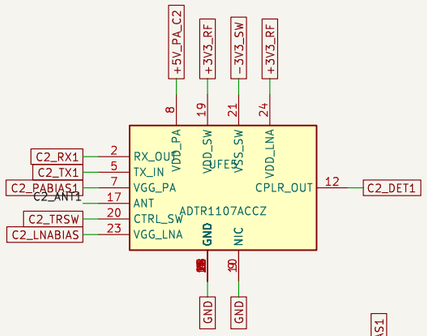


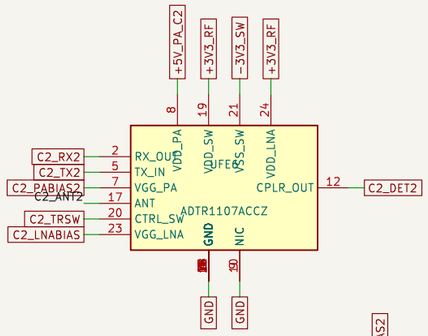


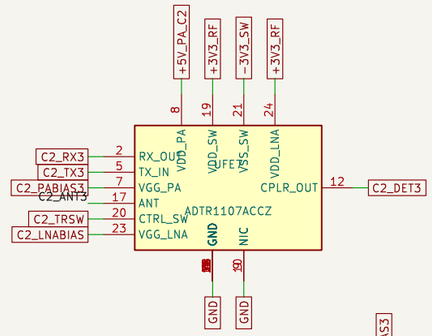
Sheet: /RF_CHAIN_2/BF_2/		
File: bf2.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 7/21

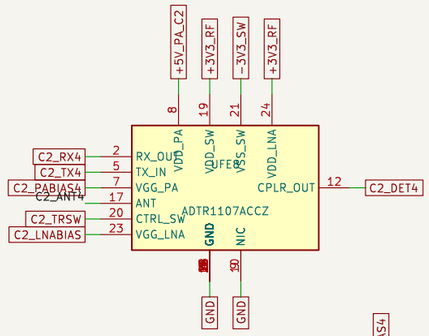
UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

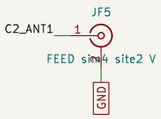
Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).

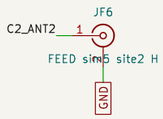


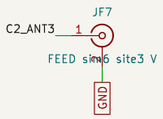


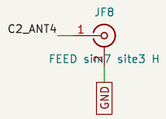


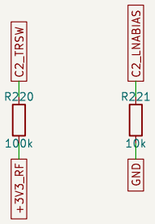












Sheet: /RF_CHAIN_2/FE_2/
File: fe2.kicad_sch

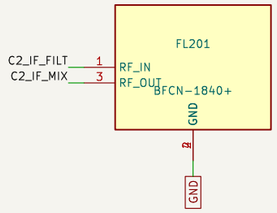
Title:

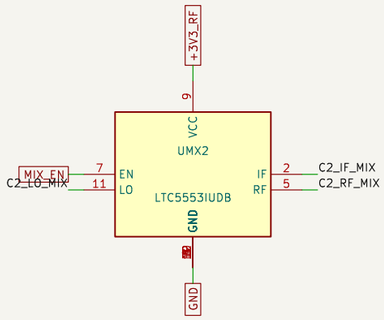
Size: A3 | Date: | **Rev:**

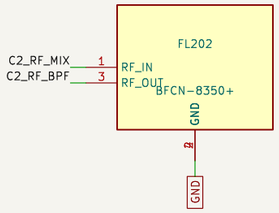
KiCad E.D.A. 9.0.9 | Id: 8/21

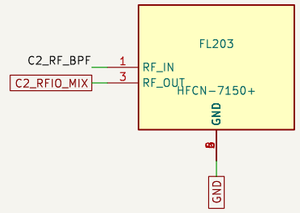
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_2/CONV_2/
File: conv2.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 9/21

RF chain 3 (topology A): N310 ch2 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF3 <-> 4x ADTR1107 <-> 4 feeds.

BF_3

File: bf3.kicad_sch

FE_3

File: fe3.kicad_sch

CONV_3

File: conv3.kicad_sch

J301

C3_IF_N310

1

SMA IF3 (N310 ch2)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_3/
File: chain3.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

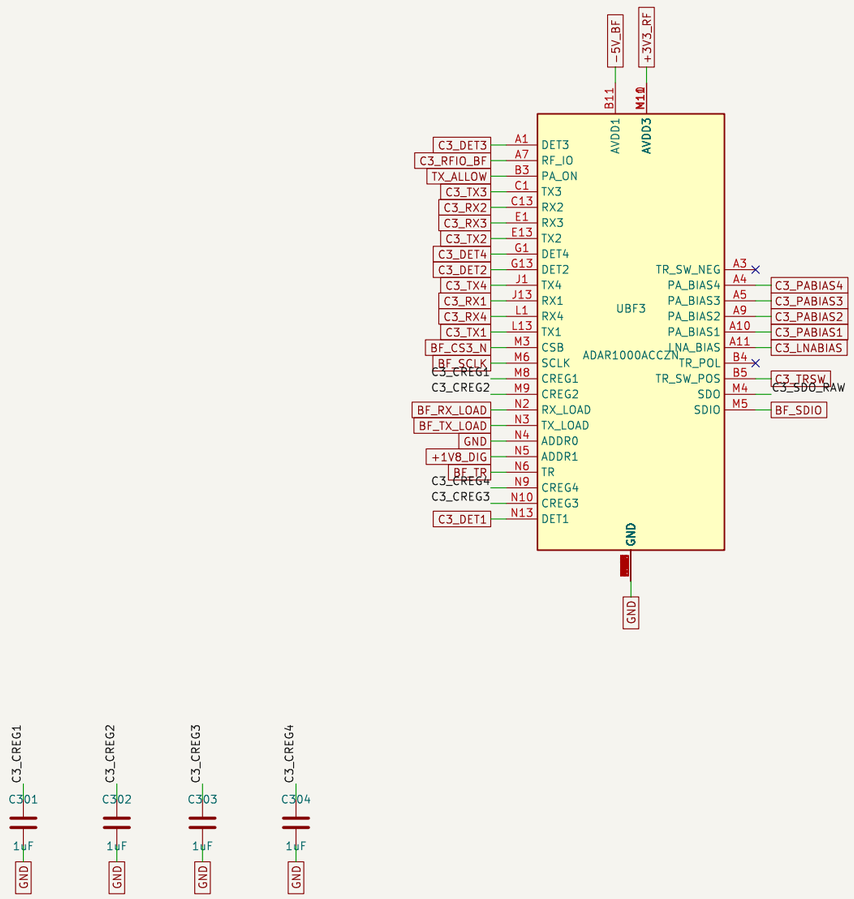
KiCad E.D.A. 9.0.9 | Id: 10/21

UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

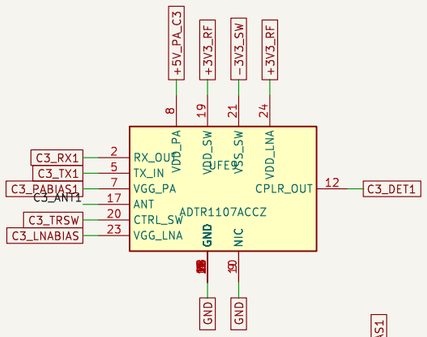
H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.

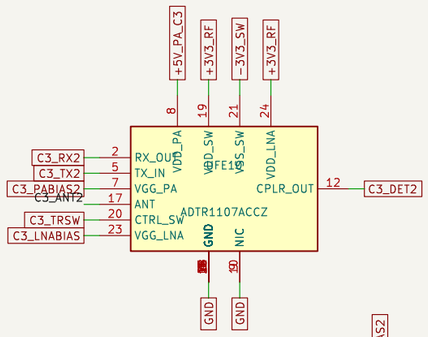


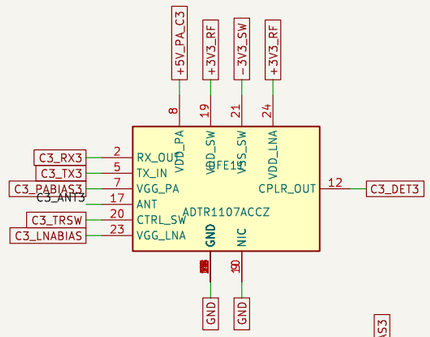
Sheet: /RF_CHAIN_3/BF_3/		
File: bf3.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 11/21

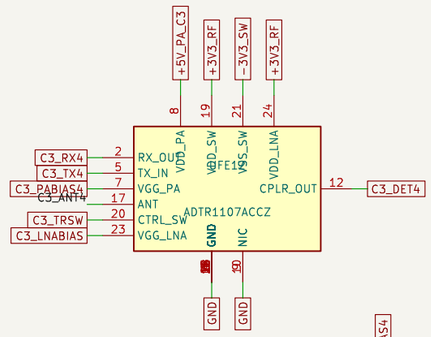
UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

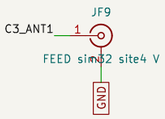
Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).

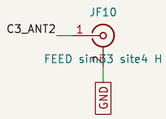


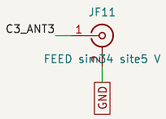




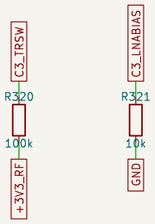












Sheet: /RF_CHAIN_3/FE_3/
File: fe3.kicad_sch

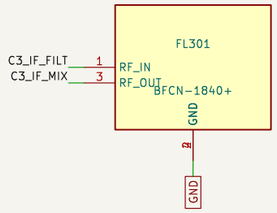
Title:

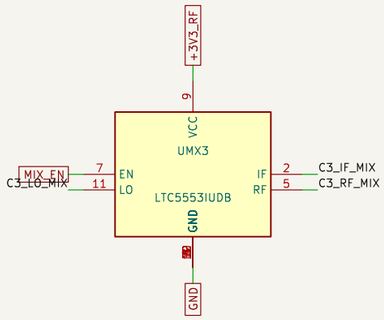
Size: A3 | Date: | **Rev:**

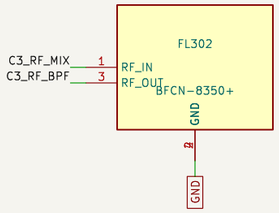
KiCad E.D.A. 9.0.9 | Id: 12/21

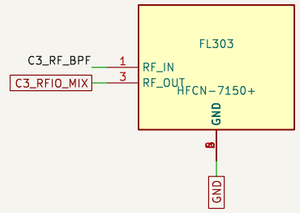
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_3/CONV_3/
File: conv3.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.9 | Id: 13/21

RF chain 4 (topology A): N310 ch3 IF 1.8 GHz <-> LTC5553 <-> 8.4 GHz <-> cal coupler <-> ADAR1000 UBF4 <-> 4x ADTR1107 <-> 4 feeds.

BF_4

File: bf4.kicad_sch

FE_4

File: fe4.kicad_sch

CONV_4

File: conv4.kicad_sch

J401

C4_IF_N310

1

SMA IF4 (N310 ch3)

GND

SMA footprint (edge-launch vs vertical) UNRESOLVED.

Sheet: /RF_CHAIN_4/
File: chain4.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

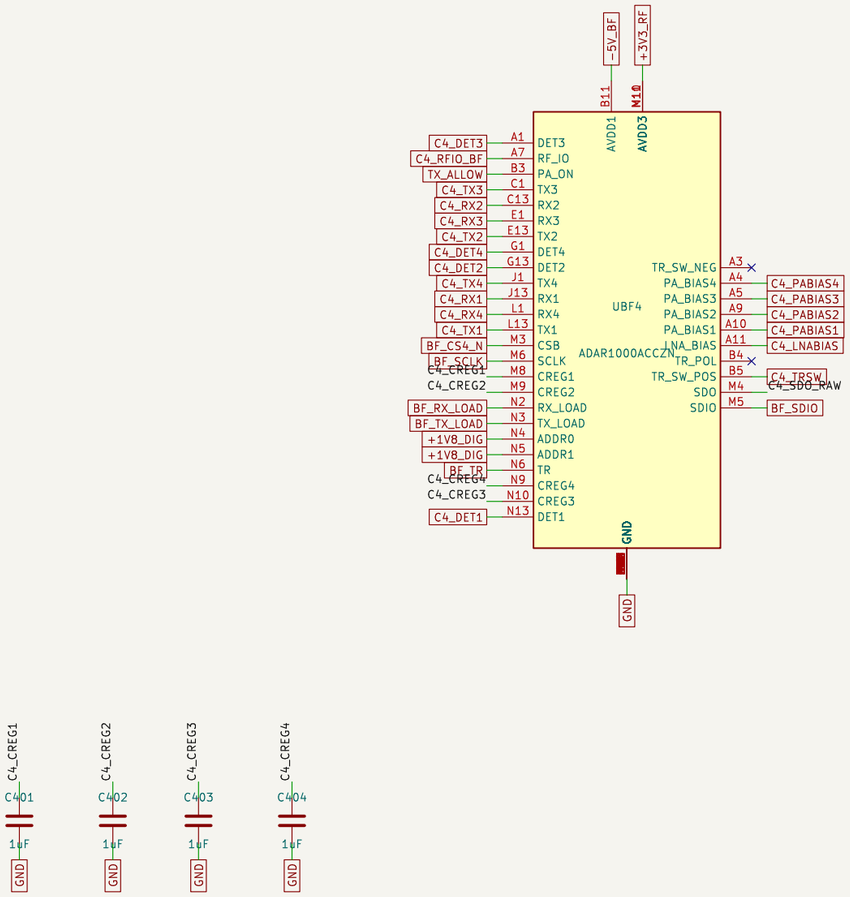
KiCad E.D.A. 9.0.9 | Id: 14/21

UNRESOLVED: AVDD3 / AVDD1 bypass capacitor values are not given in the ADAR1000 datasheet -> take from ADAR1000-EVALZ schematic.

PA_BIASx = 0 V at power-up (DS p41): PA drain stays OFF until firmware sets PA_BIAS OFF = -1.75 V (ADTR1107 VGG_PA abs max -2 V).

SDO errata (DS p47-49): non-addressed SDO not tristated -> read back one chip at a time (separate CSB).

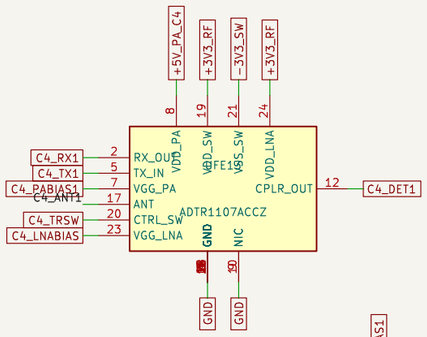
H2-B GATE-BIAS LIMIT (safety): EXT_PAx_BIAS_OFF/ON and EXT_LNA_BIAS_OFF/ON must stay <= 0x5D (~-1.75 V nominal, V = -code x 4.8/255). NEVER use ADAR1000 Rev B p45 values 0x85 (~-2.5 V) or 0x68 (~-1.96 V): ADTR1107 Rev C abs max VGG_PA -2..0 V, VGG_LNA -2..+0.2 V. Measure PA_BIASx with VDD_PA OFF before enabling the drain.

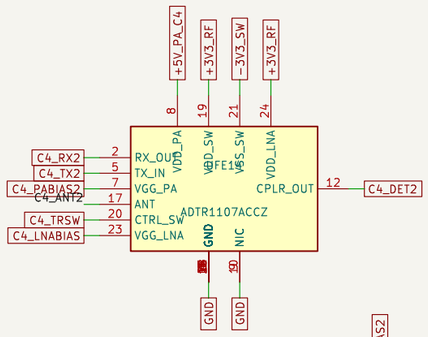


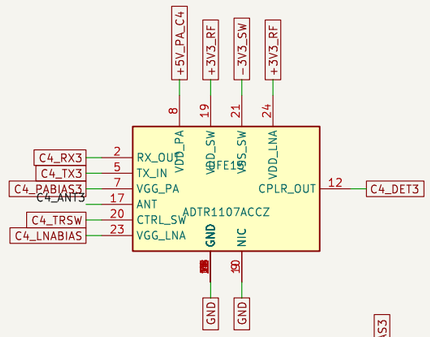
Sheet: /RF_CHAIN_4/BF_4/		
File: bf4.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 15/21

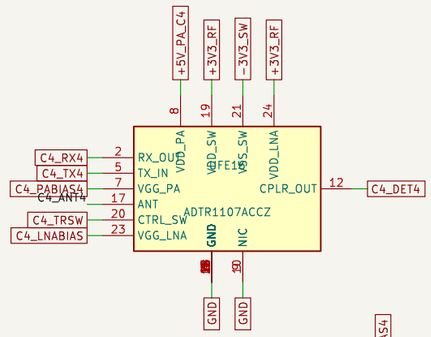
UNRESOLVED: VDD_PA / VDD_LNA / VDD_SW / VSS_SW bypass values not in ADTR1107 Rev C (Fig 89 shows only 10 nF on VGG_PA).

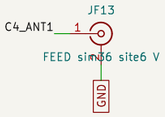
Feed ownership = H1_FEED_POLARIZATION_MAP.csv (topology A). LNA kept powered in TX and pinched via LNA_BIAS (Stingray style) — confirm with ADI (ADTR1107 TX sequence lists VDD_LNA = 0 V).

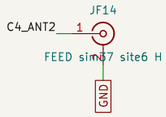


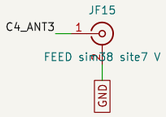


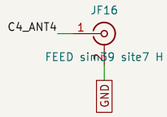


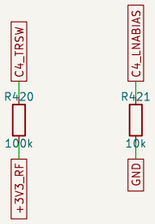








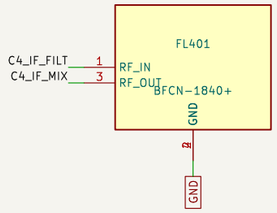


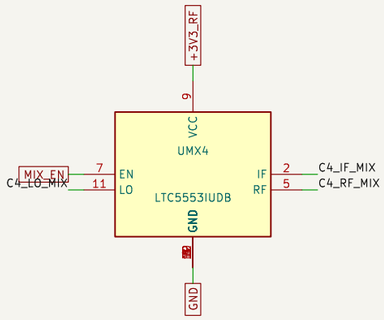


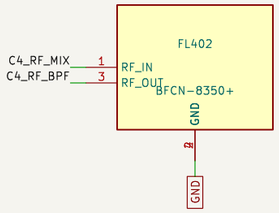
Sheet: /RF_CHAIN_4/FE_4/		
File: fe4.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 16/21

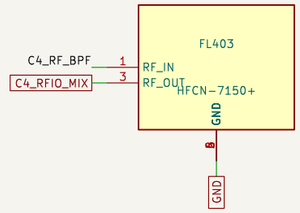
IF 1.75–1.85 GHz: N310 <–> DC block <–> BFCN–1840+ <–> LTC5553 IF (single–ended 50 ohm, no balun).

RF: LTC5553 RF <–> BFCN–8350+ <–> HFCN–7150+ <–> cal coupler (cal sheet) <–> ADAR1000 RF_IO. No DC blocks: RF_IO and LTC5553 RF are both DC–grounded.

















Sheet: /RF_CHAIN_4/CONV_4/		
File: conv4.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 17/21

UNRESOLVED: loop filter (CPOUT -> VTUNE, RS_SW) needs ADIsimPLL design; pins intentionally left unconnected.

ADF4371 RF8 is ENABLED at power-up by register default (REG0025 RF_EN=1): CE held low by pull-down until firmware programs the part.

LO path: ADF4371 RF8P 6.6 GHz -> AVA-183A+ -> 3 dB pad -> EP4RKU+ 4-way -> LTC5553 LO x4 (C1_LO..C4_LO). Equal-length routing required.

H2-B: loop filter C509/R507/C510/R508/C511 = DERIVED 3rd-order passive (Kv 50 MHz/V nominal, Icp 1.75 mA, N 660, fPFD 10 MHz; 98 kHz BW, 46 deg PM; tools/loop_filter.py) - confirm in ADIsimPLL. RS_SW (fastlock) UNRESOLVED. AVA-183A+ bias L505/C512-C515 per Mini-Circuits Fig 2 (NPO 0402, L1 wire-wound 0805): ML1-ML7 microstrips + pads 1.6 to GND per PL-328.

REF_10M
C503
100nF
PLL_REFP
C504
100nF
PLL_REFN
U501
ADF4371BCCZ
PLL_VTUNE
PLL_CS_N
BF_SCLK
PLL_CE
PLL_REFP
PLL_REFN
PLL_CP
PLL_RF8P
PLL_RF8N
PLL_AUXP
PLL_AUXN
BF_SDIO
PLL_MUXOUT_RAW
GND
C501
1uF
C502
1uF
C509
120pF
R507
5.6k
C510
820pF
R508
2.2k
C511
22pF
L501
7.5nH
L502
7.5nH
C505
10pF
LO_AMP_IN
#FLG501
PWR_FLAG
C506
10pF
R502
49.9
C512
0.1uF
LO_AMP_VD1
L505
3.3nH
U502
AVA-183A+
LO_AMP_OUT
C513
1nF
C514
18pF
C515
5.6pF
R504
294
R505
17.4
LO_SPLIT_IN
R506
294
R501
1k
L503
7.5nH
L504
7.5nH
C507
10pF
LO_MON
C508
10pF
PLL_AUXN_T
R503
49.9
U503
EP4RKU+
C1_LO
C2_LO
C3_LO
C4_LO
J501
SMA LO_MON (RFAUX8)

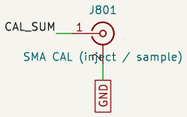
Sheet: /LO_SYNTH/
File: lo.kicad_sch

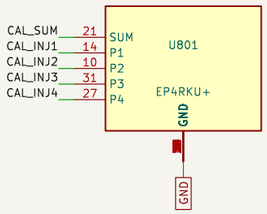
Title:

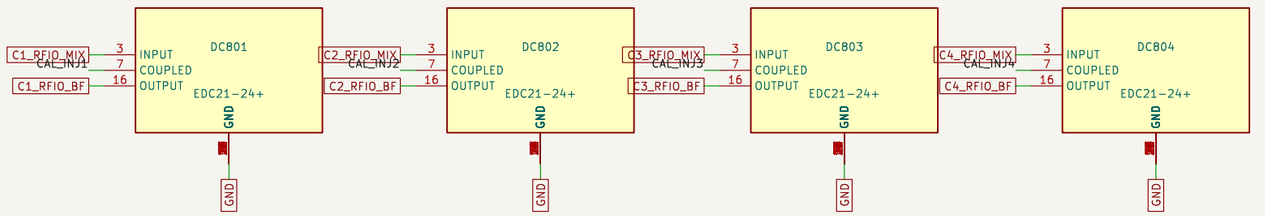
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.9 | Id: 18/21

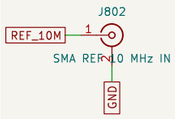
Calibration: N310 relative channel phase is random per start/retune -> run CAL-port RX/TX loopback after every N310 init or LO retune; coupler + EP4RKU+ paths must be phase-matched in layout and characterised once (VNA).

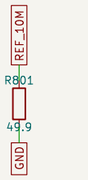
Per-element: ADTR1107 CPLR_OUT -> ADAR1000 DETx (power only, -20..+10 dBm). Element phase cal needs a near-field/OTA probe in a shielded setup (no OTA at 8.4 GHz).

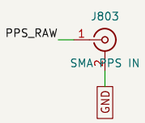




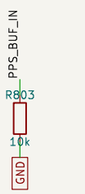


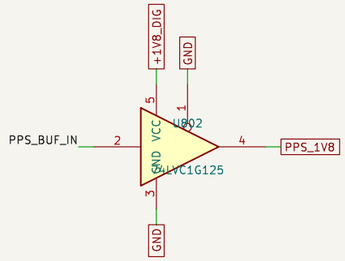












Sheet: /CAL_AND_TEST_IO/		
File: cal.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 19/21

Power-up order (ADAR1000 Rev B p44–45 + ADTR1107 Rev C p24): +1V8_DIG/+3V3_AUX (always on) -> MCU -> +3V3_RF (VDD_SW, VDD_LNA, AVDD3) -> −5V_BF & −3V3_SW (PWR_EN_NEG) -> program PA_BIAS OFF = −1.75 V, CTRL_SW = RX -> +5V_PA rail -> PA_DRAIN_EN (hardware AND) -> raise VGG_PA for 220 mA.

Power-down: TX off (BF_TR low) -> PA_BIAS to −1.75 V -> PA_DRAIN_EN low -> negative rails -> +3V3_RF. A watchdog or NEG_OK fault removes the PA drain in hardware.

All regulator blocks are UNRESOLVED placeholders (TBD_*), not parts: topology, inductors, compensation and feedback values are not designed yet.

H2-B PA drain switches U621–U624 (TI TPS22B10 SLVSDH0C): QOD tied to VOUT (internal RPD 250–400 ohm at 5 V -> fast drain discharge on disable); CT = 2200 pF (TI test condition; tR ~0.5–0.7 ms); CIN = 1 uF per switch (TI recommended). EN/UVLO VENR 1.13–1.30 V < 1.8 V logic high.

#FLG601
#FLG602
#FLG603
J601
VIN_12V
GND
VIN 12V bench
R602
10m
+3V3_RF_SRC
+3V3_RF
U601
U602
U603
U604
U605
U606
U607
U608
UNRESOLVED_REG
UNRESOLVED_NEG_REG
UNRESOLVED_NEG_MON
PWR_EN_3V3
PWR_EN_5VPA
PWR_EN_LO
PWR_EN_NEG
+3V3_RF_SRC
PG_3V3RF
+5V_PA
+1V8_DIG
+3V3_LO
+5V_LO
+3V3_AUX
−5V_BF
−3V3_SW
NEG_OK
#FLG611
#FLG612
#FLG613
#FLG614
R611
R612
R613
R614
10m
+5V_PA_IN_C1
+5V_PA_IN_C2
+5V_PA_IN_C3
+5V_PA_IN_C4
C625
C626
C627
C628
1uF
U624
U625
U626
PA_DRAIN_EN
PA_SW1_CT
PA_SW2_CT
PA_SW3_CT
PA_SW4_CT
+5V_PA_C1
+5V_PA_C2
+5V_PA_C3
+5V_PA_C4
TPS22B10DRV
C621
C622
C623
C624
2200pF
R601
1M
U610
U611
U612
U613
U614
INA226
+3V3_RF
+3V3_RF_SRC
I2C3_SDA
I2C3_SCL
+5V_PA
INA_ALERT_N
#FLG620
PCA_VREF2
R603
200k
R604
4.7k
R605
4.7k
R606
10k
I2C3_SCL
I2C3_SDA
INA_ALERT_N
U615
PCA9306
I2C_SCL
I2C_SDA

Sheet: /POWER_SEQUENCING/
File: power.kicad_sch

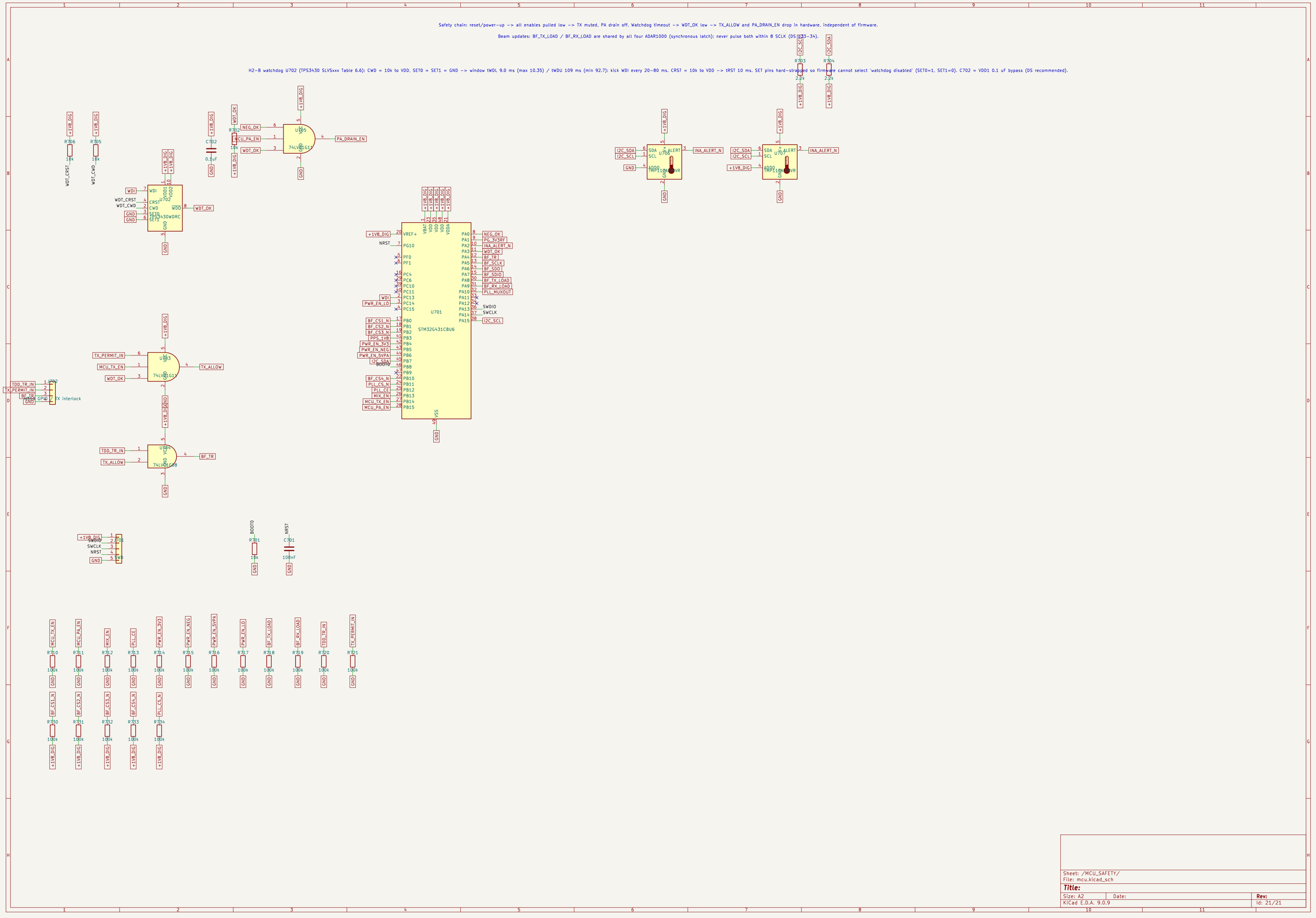
Title:

Size: A2 | Date: | **Rev:**
KiCad E.D.A. 9.0.9 | Id: 20/21

Safety chain: reset/power-up -> all enables pulled low -> TX muted, PA drain off. Watchdog timeout -> WDT_OK low -> TX_ALLOW and PA_DRAIN_EN drop in hardware, independent of firmware.

Beam updates: BF_TX_LOAD / BF_RX_LOAD are shared by all four ADAR1000 (synchronous latch); never pulse both within 8 SCLK (DS 33-34).

H2-B watchdog U702 (TPS3430 SLVSxxx Table 6.6): CWD = 10k to VDD, SET0 = SET1 = GND -> window tWDL 9.0 ms (max 10.35) / tWDU 109 ms (min 92.7); kick WDI every 20-80 ms. CRST = 10k to VDD -> tRST 10 ms. SET pins hard-strapped so firmware cannot select 'watchdog disabled' (SET0=1, SET1=0). C702 = VDD1 0.1 uF bypass (DS recommended).



Sheet: /MCU_SAFETY/	
File: mcu.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 9.0.9	**Rev:** Id: 21/21